100% Excellent!

PHY201 Exam on chapter(s): 1, 2, 3

Name: [redacted]

These questions assess your understanding of the material from Chapters 1, 2, 3 of Silberberg. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must explain why you chose your answer. Remember to report the units and correct number of significant figures in your answers.

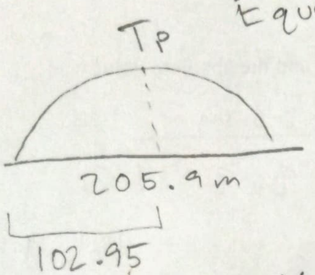
1. Jennifer aims her arrow towards a target located 205.9 m away. The target's bull's eye aligns with the height from which Jennifer releases the arrow. The arrow is launched at an angle with an initial speed of 86.5 m/s. Find the angle at which Jennifer launched the arrow.

ANSWER: $7.82°$ above horizon ✓

$\star \ \sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$

Equate T_p $\quad \Delta x_1 = V_{0x} t_p + \cancel{\frac{1}{2} a t^2}$

good!



$\frac{\Delta x_p}{V_0 \cos\theta} = t_p$ $\quad V_{fy} = 0 = V_{0y} + at$

$\Rightarrow V_{0y} = gt$

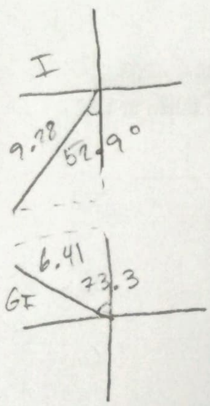
$\frac{x_p}{V_0\cos\theta} = \frac{V_0\sin\theta}{g}$ $\quad \frac{V_{0y}}{g} = t_p$

$\frac{x_p g}{V_0^2} = \sin\theta\cos\theta = \frac{1}{2}\sin 2\theta \Rightarrow \theta = \frac{1}{2}\sin^{-1}\left(\frac{2x_1 g}{V_0^2}\right) = 7.8226°$

OR just use $R = \frac{V_0^2 \sin 2\theta}{g}$

2. Tyler is sitting in his folding chair on the beach, watching his friend Ian as he rides in a catamaran moving at a velocity of 9.28 m/s at an angle of 52.9° west of south. Ian then passes his sister, Grace, who is riding in a trimaran, which, from Ian's perspective, is travelling at a velocity of 6.41 m/s at an angle of 73.3° west of north. From Tyler's perspective, what is the magnitude of the velocity of Grace on the trimaran?

ANSWER: 14.052 m/s ✓



$V_{GI} = V_G - V_I$

$V_{GI} + V_I = V_G$

$\Sigma V_x = V_{GIx} + V_{Ix}$

$\Sigma V_x = 6.41\sin 73.3 + 9.28\sin 52.9 = 13.54122$

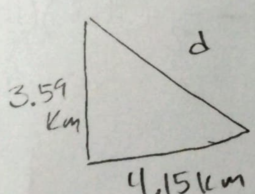
$\Sigma_y = 6.41\cos 73.3 - 9.28\cos 52.9 = -3.75579$

$V_G = \sqrt{(13.54122)^2 + (3.75579)^2} = 14.05242 \text{ m/s}$

3. Out of curiosity, you want to conduct some geographic measurements of your house and your friend Allison's. To get to Allison's, you have to walk 4.15 km west and then 3.59 km north. How far does she live from you if you drew a straight line on the map?

ANSWER: 5.487 Km ✓

$d = \sqrt{4.15^2 + 3.59^2}$



$d = 5.487$ Km

+3

✓ 4. A basketball referee tosses the ball straight up for the starting tip-off. What is the minimum velocity at which a basketball player must leave the ground in order to rise 3.17 m above the floor and make contact with the ball?

ANSWER: 7.882 m/s

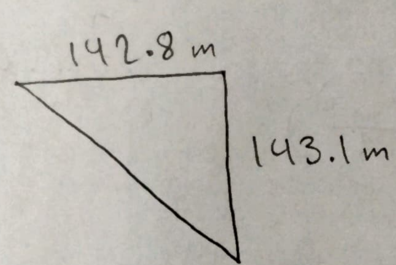
$$h = \frac{V_o^2}{2g}$$

$$\sqrt{2gh} = V_o$$

$$\sqrt{2(9.8)(3.17)} = 7.882 \text{ m/s}$$

✓ 5. A man walks 142.8 m east and then 143.1 m south. What is the difference between his travelled distance and the absolute value of his displacement?

ANSWER: 83.738 m



$$d = \sqrt{142.8^2 + 143.1^2} = 202.162$$

$$trav = 142.8\text{ m} + 143.1\text{ m} = 285.9$$

$$285.9 - 202.162 = 83.738 \text{ m}$$

✓ 6. Gabriel is walking on a bridge between two mountain peaks, when curiosity suddenly overcomes him and he throws a beanbag straight down from the bridge to the distant vale below. If the beanbag falls 36 m in 1.7 s, then how far will it have fallen in 10.2 s?

ANSWER: 640.83 m

$V_o = ?$
$Y_o = 0$

$V_1 = ?$
$Y_1 = 36$
$t_1 = 1.7$

$V_2 = ?$
$Y_2 = ?$
$t_2 = 10.2$

$$\Delta y_1 = V_o t_1 + \frac{1}{2} a t_1^2$$

$$\frac{\Delta y_1 - \frac{1}{2} a t_1^2}{t_1} = V_o = -12.846\ldots$$

$$\Delta y_2 = \left(\frac{\Delta y_1 - \frac{1}{2} a t_1^2}{t_1}\right) t_2 + \frac{1}{2} a t_2^2$$

$$\Delta y_2 = \left(\frac{-36\text{m} + \frac{1}{2}(9.8)(1.7)^2}{1.7}\right) 10.2 - \frac{1}{2}(9.8)(10.2)^2$$

$$= -640.83 \text{ m}$$

+3

7. A small airplane flies in a straight line at a speed of 184 km/h. How long does it take the plane to fly 105 leagues? Report your answer in minutes. An inch is 2.54 cm, a mile is 2640 cubits, a league is 3 miles, and a cubit is 2 feet.

ANSWER: 165.29 min ✓

~~184 km | 1000 m / 1 km | 100 cm / 1 m | / 2.54 cm~~

good!

$$105 \text{ leagues} \left| \frac{3 \text{ miles}}{1 \text{ league}} \right| \frac{2640 \text{ cubits}}{1 \text{ mile}} \left| \frac{2 \text{ ft}}{1 \text{ cubit}} \right| \frac{12 \text{ in}}{1 \text{ ft}} \left| \frac{2.54 \text{ cm}}{1 \text{ in}} \right| \frac{1 \text{ m}}{100 \text{ cm}} \left| \frac{1 \text{ km}}{1000 \text{ m}} \right.$$

$= 506.943$ km

$v = \frac{\Delta x}{t} \Rightarrow t = \frac{\Delta x}{v}$

$\frac{184 \text{ km}}{1 \text{ hr}} \left| \frac{1 \text{ hr}}{60 \text{ min}} \right. = 3.067 \text{ km/min}$

$= \frac{506.943 \text{ km}}{3.067 \text{ km/m}} = 165.29$

8. Leah drops a coconut off the edge of a cliff and measures it striking the ground with a vertical impact velocity of 73.1 m/s. Determine the height of the cliff. Report your answer in feet, recalling that 3.28 ft = 1 m.

ANSWER: 894.237 ft ✓

$v_{fy}^2 = \cancel{v_{oy}^2} + 2a(\Delta y)$

$\frac{v_{fy}^2}{2a} = \Delta y$

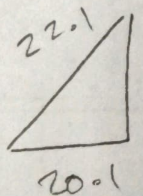
$\frac{(73.1 \text{ m/s})^2}{2(9.8)} = 272.633 \text{ m}$

$272.633 \text{ m} \left| \frac{3.28 \text{ ft}}{1 \text{ m}} \right.$

$= 894.237$ ft

9. The magnitude of a velocity vector is 22.1 m/s and its x component's magnitude is 20.1 m/s. What is the magnitude of its y component?

ANSWER: 9.187 m/s ✓



$v = \sqrt{v_x^2 + v_y^2}$

$v^2 = v_x^2 + v_y^2$

$v^2 - v_x^2 = v_y^2$

$\sqrt{v^2 - v_x^2} = v_y = 9.187 \text{ m/s}$

+3

Answers for exam 1.capitalize() (Chapters: 1, 2, 3)

1. 7.8° above the horizontal
2. 14.1 m/s
3. 5.5 km
4. 7.88 m/s
5. 83.7 m
6. 641 m
7. 165 minutes
8. 894 ft
9. 9.2 m/s